

Space age

The sky up above just can't be an unlimited litterbox. Find out what mankind is doing to control space debris

Rise of LoL

This is the story of how one video game turned into a multimillion dollar enterprise and professional sport: <http://dgit.in/RpjBjf>

Anti-Snooping Devices

The internet is seeing a dramatic shift towards anonymity.

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Remember in Dan Brown's novel *Digital Fortress*, when NSA agent Susan Fletcher is trying to figure out why their famed supercomputer ('TRANSLTR') is taking forever to decrypt a seemingly innocuous code? Turns out the 'code' was actually a worm designed to bring down the supercomputer, written by an ex-NSA agent who was unhappy with the NSA being privy to the personal communication of millions of people over the world.

Turns out, the NSA are huge Dan Brown fans.

In an eerily similar series of events, the NSA (National Security Agency, USA) was found to have conducted mass electronic surveillance across internet services such as Google, Facebook, Yahoo, Amazon etc. The real-life analogue to Dan Brown's tragic hero, played by a 29 year old ex-NSA agent, Edward Snowden, stated that the reason behind leaking the documents which made this big reveal was that he didn't wish to live in a society that indulges in such unethical practices. Snowden fled the US, seeking political asylum in a number of countries, as his home country launched an all-out man-hunt to track him down.

A wake up call

In response to these leaks last year, several experts began stressing the importance of securing your personal data and communications from eternal surveillance, chiefly by encrypting it. Encrypted data can only be read by someone who has a 'key' to it - a password that unlocks

the true contents of the message. While encryption isn't a one-stop solution, it's better than nothing. However, as is always the case, increased security comes at the cost performance overheads. Encryption isn't fool-proof, nor can everything be encrypted. Take WhatsApp. It stores chats in an encrypted database on the phone's local storage. As a proof of concept, a hacker gained access to the encryption keys and was able to read all the user's chat messages in cleartext. (<http://dgit.in/WhatsAppHack>) In theory, this



The Boeing Black will destroy all data and software if it is forced open.

means any app installed on your phone can read/modify/upload these chats to an external server and wreck havoc with your confidential data.

The vaults of the new age

A number of security-conscious companies and individuals expressed concern over the heightened surveillance, and nearly a year later, we're seeing the fruits of their effort. Secure smartphones such as Silent Circle's **BlackPhone** and **Boeing's**

Black, as well as encryption-focussed hardware are making themselves seen and heard in the market. These devices usually feature secure, encrypted communication channels as well as features that make the device tamper-proof. Koolspan has a product called the **TrustedChip**, a chip that can be plugged into the microSD card slot of an Android phone, which features an encryption engine that encrypts all data transactions on the device.

The Boing Black

Boeing, the American aircraft manufacturer, recently unveiled the Boeing Black, an Android based smartphone aimed at industries requiring a high level of secrecy, such as defence establishments and government agencies. Its specs are fairly standard for a mid-range smartphone - a 4.3" qHD display with a resolution of 960x540, a dual-core 1.2GHz ARM v9 CPU, a 1590mAh battery and a dual-SIM supporting both GSM and CDMA networks. What's special about it, is that it runs a custom version of Android that includes a "Hardware Encryption Engine" that encrypts the local flash storage, as well as any data transmitted from the device. The back panel of the device can be customized to feature a solar-charger, a biometric scanner and even satellite transceivers. A feature called the "Hardware Root of Trust" verifies the authenticity of any software/apps that are installed on the device through a set of digital signatures, while a Secure Boot feature ensures that only secure and approved system images can be booted on the device. If the system image is modified in any manner by an unauthorized source, the device will simply refuse to boot. The real clincher, though, is its self-destruct